Asif Khan

Staff Scientist in AI for Medicine

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Research Profile

Machine learning researcher developing representation learning, generative models, and foundation models for population-scale clinical data and cancer biology. My work spans longitudinal electronic health records, calibrated pan-cancer risk stratification, domain-adaptive pretraining across healthcare systems, drug and CRISPR perturbation modeling, and geometric approaches to representation learning and interpretability. I work across methodological, clinical, and translational collaborations to build models that are scalable, robust, and useful in practice.

Current Appointments and Affiliations

Staff Scientist Current
Department of Systems Biology, Harvard Medical School Boston, MA

Visiting Postdoctoral Fellow Current
Institute for Systems Biology, Hadlock Lab (Jennifer Hadlock) Seattle, WA

Research Associate Current
Cancer Early Detection and Diagnostics Clinic, Massachusetts General Hospital Boston, MA

Research Affiliate Current
Ludwig Center at Harvard; Broad Institute of MIT and Harvard Boston, MA

Education

PhD in Machine Learning 2019–2023
University of Edinburgh; advisor: Amos Storkey Edinburgh, UK

Dissertation: *Geometry for Deep Representation Learning.*

MSc in Computer Science, grade 1.1 (excellent) 2017–2019
University of Bonn Bonn, Germany

Research in machine learning, knowledge graphs, and generative models for voice morphing.

Research Experience

Harvard Medical School 2024–present
Postdoctoral Fellow; Staff Scientist (Chris Sander and Debora Marks) Boston, MA

- Develop generative models of longitudinal patient health states for pan-cancer risk stratification using population-scale EHR data.
- Build robust, calibrated clinical prediction systems across cancer types, institutions, and patient populations.
- Model cellular responses to drug and CRISPR perturbations to support combination-therapy design.
- Lead multidisciplinary collaborations spanning machine learning, clinical oncology, health data science, and cancer biology.

University of Edinburgh 2019–2023
PhD Researcher in Machine Learning (Amos Storkey) Edinburgh, UK

- Developed geometric and topological methods for understanding deep representation learning.
- Designed Hamiltonian latent operators for content and motion disentanglement and methods for robust generative models.
- Developed multi-resolution structural views for representation learning on non-local graphs.

Huawei Noah's Ark Lab 2021
Research Scientist Intern London, UK

- Developed combinatorial Bayesian optimization for automated antibody design with uncertainty quantification.

Sony Europe 2019
Research Intern Stuttgart, Germany

- Developed generative models for unsupervised cross-domain speech-to-speech conversion.

KAUST Bio-Ontology Research Group 2016–2017
Research Assistant Jeddah, Saudi Arabia

- Developed ontology-aware neural networks and neuro-symbolic methods for biological knowledge graphs.

Selected Publications

1. **A. Khan**, D. T. Forster, M. Harsh, C. Zheng, E. T. Warner, D. Ritter, A. Chang, Q. Wei, T. K. Sorensen, D. S. Marks, L. V. Sequist, J. J. Hadlock, N. R. Fillmore, and C. Sander. Identifying cohorts at elevated risk of cancers using generative modeling of patient health states. *medRxiv*, 2026.
2. **A. Khan**. Capacity and redundancy trade-offs in multi-task learning. *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.
3. **A. Khan**. The geometrical and topological signature of transformers. *ICLR Workshop on Geometry-grounded Representation Learning and Generative Modeling*, 2026.
4. Y. Sun et al., including **A. Khan**. AI proteomics: from protein identification to virtual cells. *Nature Methods*, 2026.
5. L. M. Luque, **A. Khan**, G. Torrisi, T. D. Green, et al. Identifying tissue states by spatial protein patterns related to chemotherapy response in triple-negative breast cancer. *eLife*, 2026.
6. **A. Khan***, C. Zheng*, D. Ritter*, D. S. Marks, N. V. Do, N. R. Fillmore, and C. Sander. Pancreatic cancer risk prediction using deep sequential modeling of longitudinal diagnostic and medication records. *Cell Reports Medicine*, 2025. (*Joint first authors)
7. **A. Khan***, D. Ritter*, C. Zheng, D. T. Forster, et al. Multi-cancer risk prediction using transformers trained on large-scale longitudinal EHR data. *NeurIPS Workshop on Learning from Time Series for Health*, 2025. (*Joint first authors)
8. **A. Khan**, D. T. Forster, M. Harsh, D. Ritter, et al. Position: AI-driven risk stratification is essential for affordable early detection of cancer. *NeurIPS Workshop on GenAI for Health*, 2025.
9. **A. Khan**, G. Torrisi, L. M. Luque, et al. Batch-effect invariant graph neural networks for predicting chemotherapy response in triple-negative breast cancer patients. *ICML Workshop on Machine Learning for Life and Material Sciences*, 2024.
10. **A. Khan** and A. Storkey. Adversarial robustness of VAEs through the lens of local geometry. *AISTATS*, 2023.
11. **A. Khan***, A. I. Cowen-Rivers*, A. Grosnit, et al. Toward real-world automated antibody design with combinatorial Bayesian optimization. *Cell Reports Methods*, 2023. (*Joint first authors)
12. **A. Khan** and A. Storkey. HALO: Hamiltonian latent operators for content and motion disentanglement in image sequences. *NeurIPS*, 2022.
13. M. A. Khan*, A. Kristiadi*, D. Lukovnikov, J. Lehmann, and A. Fischer. Incorporating literals into knowledge graph embeddings. *ISWC*, 2019. (*Joint first authors)
14. M. Kulmanov, M. A. Khan, and R. Hoehndorf. DeepGO: Predicting protein functions from sequence and interactions using a deep ontology-aware classifier. *Bioinformatics*, 2017.

Complete publication list: mdasifkhan.github.io/publications

Invited and Selected Talks

New England Computational Biology Symposium (NECB) 2026
Selected talk: Pan-cancer risk assessment with an EHR foundation model that predicts what happens next and when Microsoft Research New England

The Cancer Prevention Research Conference 2026
Flash talk: Pan-cancer risk stratification via a foundation model of patient health histories Atlanta, GA

Cancer Early Detection and Diagnostics Clinic 2026
Pan-cancer risk assessment for early detection and prevention from real-world longitudinal clinical records Massachusetts General Hospital, Boston

AI in Cancer Research Symposium 2026
Invited tutorial: AI in Oncology Ludwig Center at Harvard

Hadlock Lab Seminar 2025
Cancer risk prediction from large-scale electronic health records Institute for Systems Biology

MIT-MGB AI Cures 2025
Representation learning on longitudinal health trajectories using LLMs for disease risk assessment Cambridge, MA

Funding and Awards

Best Poster Award, The Early Detection of Cancer Conference 2026

Best Poster Award, The Cancer Prevention Research Conference 2026

Fund for Innovation in Cancer Informatics 2025–2027
Fellowship for multi-risk cancer prediction using longitudinal EHRs

DAAD Postdoc-NeT-AI 2024
AI for Science Fellow

NeurIPS Scholar Award 2022

PhD Scholarship, University of Edinburgh 2019–2023

Academic Service

Journal reviewer: Journal of Machine Learning Research (JMLR).
Conference reviewer: NeurIPS 2026; ICML 2026 (Technical Reviewer: Gold); NeurIPS 2025; ICML 2025 (Outstanding Reviewer); NeurIPS 2024 (Top Reviewer); ICML 2024 (Best Reviewer); NeurIPS 2023 (Top Reviewer); ICML 2023; AISTATS 2022–2023 (Top Reviewer); NeurIPS 2022 (Top Reviewer); ICLR 2022 (Highlighted Reviewer).
Workshop reviewing and program committees: CAO and GRaM at ICLR 2026; TS4Health at NeurIPS 2025; ML4LMS at ICML 2024; SynS & ML at ICML 2023; ML4PS at NeurIPS 2021–2023.

Teaching

University of Edinburgh 2019–2023
Tutor and marker Edinburgh, UK

Probabilistic Modelling and Reasoning; Machine Learning Practical; Introductory Applied Machine Learning; Data Mining and Exploration.

University of Bonn 2017–2019
Teaching Assistant, Knowledge Graph Analysis Bonn, Germany

Delivered tutorials and prepared theoretical and programming exercises and assessments.

Student Mentoring

Sreevatsa Nukala	Master's, Harvard T.H. Chan School of Public Health	2025–2026
Abdullah Kuzeiz	Master's, Harvard-MIT Health Sciences and Technology	2025
Sophie Sharum	Undergraduate, Harvard University	2025
Nicole Xu	Undergraduate, MIT	2024
Michael D. Dennison	Undergraduate, MIT	2024

Research Collaborations

Chris Sander (Harvard Medical School / Ludwig / Broad); Debora Marks (Harvard Medical School); Jennifer Hadlock (Institute for Systems Biology); Lecia Sequist (Mass General Brigham Cancer Institute); Nathanael Fillmore (VA Boston / Harvard Medical School); Linus Schumacher (University of Edinburgh).

Technical Expertise

Methods: representation learning, generative models, self-supervised learning, geometric deep learning, uncertainty quantification, calibration, large language models.
Applications: longitudinal EHR modeling, cancer risk stratification, spatial proteomics, drug and CRISPR perturbation modeling, computational drug discovery.
Tools: Python, PyTorch, NumPy, scikit-learn, SQL, Bash, Git, Hugging Face, GCP.